

Paper 0.25 - Toolkit for the First Section

CQE-paper-00.25

CQECMPLX Formal Paper Series

Review-facing PDF built 2026-06-13

Construction Basis

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-00.25.md

Paper 0.25 - Toolkit for the First Section

Purpose

Paper 0.25 describes the tools a reader may use while reviewing the first section of papers. It is a supplement, not a substitute for the scientific papers. Its job is to make the tests reproducible in multiple media.

Tool Classes

The first section uses four visible tool classes.

```
formal paper source
executable verifier
receipt or proof ledger
analog exposure route
```

A claim may be reviewed entirely from the paper if the proof is clear. The tools exist so the same claim can also be replayed, inspected, or challenged.

Digital Tools

The digital layer includes:

- paper source files,
- verifier scripts,
- JSON receipts,
- kernel manifests,
- lib-forge package bindings,
- lookup caches,
- archive indexes.

These should be used to answer concrete review questions:

```
What exactly is claimed?
What input does the test use?
What output counts as success?
What output counts as failure?
Where is the receipt?
Can the result be regenerated?
What remains an obligation?
```

Analog Tools

The analog layer exists to expose the same state in ordinary visible form:

- a center mark,
- left/right boundary marks,
- a transform arrow,
- a receipt line,
- a proof/obligation split,
- a next-state page.

The analog route is not a ritual. It is a way of showing that the formal claim does not depend

on hidden code. If the claim can be reduced to a center, a boundary, a transform, and a receipt, then a reviewer can inspect the structure independently of the software.

Use Rule

For each integer paper in the first section, the toolkit asks:

```
Can the claim be read as a center/boundary state?  
Can the transform be stated?  
Can the receipt be regenerated?  
Can a failure be preserved without changing the claim?  
Can the same state be exposed without the original software?
```

If the answer is no, the paper should carry an obligation rather than a proof claim.

Boundary

Paper 0.25 does not prove any mathematical or physical result. It defines the review tools available for testing the papers that follow.